

Understanding High PM2.5 Episodes in Kathmandu Through Clustering, Association Rules, and Time-Aware Classification

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Abstract

Kathmandu Valley experiences recurrent fine-particulate pollution, but meteorological conditions associated with high-PM2.5 episodes are often nonlinear and season-dependent. We analysed 7,637 hourly observations spanning 2019. Exploratory analysis was combined with standardized K-means clustering, Apriori association-rule mining and five supervised classifiers. Weather variables were discretized into dataset-relative tertiles; PM2.5 at or above 55.5 micrograms/m³ was treated as an Unhealthy-or-worse screening outcome. Prediction was evaluated with random, chronological and rolling time-series validation. Mean PM2.5 was 46.10 micrograms/m³, rising to 73.24 in winter and falling to 17.76 in monsoon. Dry-hour concentrations averaged 54.50 versus 28.29 during wet hours. Four weather regimes maximized the tested silhouette score (0.294). A cool, drier, higher-pressure and lower-wind regime had mean PM2.5 of 63.50. Apriori produced 3,753 frequent itemsets and 208 high-PM2.5 rules; the leading rule combined medium cloud cover, low temperature and low wind (support 0.021, confidence 0.599, lift 2.31). Random-split Random Forest ROC-AUC was 0.909, but the best chronological ROC-AUC was 0.711 and rolling mean AUC peaked at 0.663, revealing substantial temporal optimism. High PM2.5 co-occurred most consistently with cool, dry, high-pressure and weak-dispersion conditions. These associations are useful for hypothesis generation and screening, but one-year, single-location data and strong seasonality limit causal and predictive interpretation.

Keywords: PM2.5; Kathmandu; meteorology; association-rule mining; K-means; time-series validation

1 Introduction

Fine particulate matter (PM2.5) is a major environmental-health concern because particles small enough to penetrate deep into the respiratory system are associated with cardiopulmonary harm. The World Health Organization recommends annual and 24-hour PM2.5 guideline values of 5 and 15 micrograms/m³, respectively [1]. Kathmandu Valley combines dense urban emissions with basin topography that can inhibit dispersion. Earlier observations documented winter pollution cycles governed by local air-flow, inversion breakup and daytime mixing [2], while winter field campaigns found persistently elevated PM2.5 and shallow nocturnal boundary layers [3].

Meteorology does not create most primary emissions, but it regulates dilution, removal, chemical formation and transport. Rain scavenges particles, wind enhances ventilation, and stable high-pressure or inversion conditions can trap emissions near the surface. Recent Kathmandu research likewise reports strong seasonal effects and contributions from biomass burning, transport and boundary-layer dynamics [4, 5]. Consequently, a simple global correlation is unlikely to capture the combinations under which episodes occur.

This study asks: Which combinations of meteorological conditions are associated with high PM2.5 episodes in Kathmandu? We triangulate the answer using (i) descriptive statistics, (ii) weather-only clustering, (iii) interpretable association rules and (iv) supervised prediction evaluated with time-aware validation. The emphasis is pattern discovery rather than causal attribution.

2 Materials and methods

2.1 Dataset and preprocessing

The supplied dataset contained hourly PM2.5, ozone and meteorological observations from 1 January through 31 December 2019. After parsing timestamps and applying the project cleaning pipeline, 7,637 rows and 18 analysis fields remained. No duplicates, invalid dates or residual missing values were present. Derived hour, month and season fields used Winter (December–February), Pre-monsoon (March–May), Monsoon (June–September) and Post-monsoon (October–November).

Table 1: Analytical dataset summary.

Property	Value
Temporal coverage	1 Jan–31 Dec 2019
Observations	7,637 hourly records
Mean PM2.5	46.10 micrograms/m ³
Median / range	37 / 1–985 micrograms/m ³
Weather fields	Temperature, humidity, dew point, precipitation, pressure, cloud, vapour-pressure deficit, wind

2.2 Exploratory analysis

We summarized PM2.5 by season and wet/dry state and calculated Pearson correlations with numeric variables. Correlations were treated as descriptive because meteorological variables are mutually dependent and the hourly observations are serially correlated.

2.3 Weather-regime clustering

K-means used seven standardized meteorological inputs: temperature, relative humidity, precipitation, mean-sea-level pressure, cloud cover, vapour-pressure deficit and wind speed. PM2.5 was excluded so that clusters represented weather regimes rather than pollution-defined groups. Candidate k values from 2 to 8 were compared using inertia and silhouette score; the highest tested silhouette determined k . Cluster profiles were subsequently compared with observed PM2.5.

2.4 Association-rule mining

Temperature, relative humidity, wind speed, pressure, cloud cover and vapour-pressure deficit were converted to Low, Medium and High categories using rank-based tertiles. Rain was Wet/Dry and wind direction was grouped into cardinal sectors. PM2.5 was mapped to US EPA 24-hour AQI concentration categories [6]. Because the records are hourly, these labels are screening categories and not official daily AQI classifications. High outcomes were Unhealthy, Very Unhealthy or Hazardous (PM2.5 ≥ 55.5 micrograms/m³). Apriori used minimum support 0.02, confidence 0.35, lift 1.0 and maximum itemset length 4. Only rules with a single high-PM2.5 consequent and no PM2.5 item in the antecedent were retained.

2.5 Classification and validation

Decision Tree, Gaussian Naive Bayes, 15-neighbour k-NN, Logistic Regression and Random Forest classifiers predicted PM2.5 ≥ 55.5 from ten weather variables. We compared a stratified random 80/20 split, a chronological 80/20 split and five-fold expanding-window TimeSeriesSplit. Accuracy, precision, recall, F1 and ROC-AUC were reported. A further analysis compared weather-only, calendar-only (month and hour), and combined Random Forest inputs.

3 Results

3.1 Descriptive and seasonal patterns

PM2.5 was strongly right-skewed (mean 46.10, median 37, maximum 985 micrograms/m³). Winter had the highest seasonal mean (73.24), followed by pre-monsoon (59.43), post-monsoon (43.45) and monsoon (17.76). Dry hours averaged 54.50 compared with 28.29 in wet hours. Dew point showed the largest negative correlation with PM2.5 ($r = -0.467$), followed by temperature and cloud cover (both $r = -0.356$). Mean-sea-level pressure showed the largest positive correlation ($r = 0.392$).

Table 2: Seasonal PM2.5 summary (micrograms/m³).

Season	n	Mean	Median	Maximum
Winter	1,867	73.24	63	985
Pre-monsoon	1,954	59.43	51	985
Monsoon	2,593	17.76	15	835
Post-monsoon	1,223	43.45	38	985

The marked monsoon reduction and wet-hour contrast are consistent with particle removal by precipitation and improved seasonal ventilation, while winter accumulation agrees with prior Kathmandu observations [2, 3, 4, 5]. The isolated maximum values also indicate either extreme episodes or potential sensor/data-quality issues that deserve audit before operational use.

3.2 Meteorological regimes

Four clusters achieved the highest tested silhouette score (0.294), although the modest value indicates overlap among regimes. Cluster 0 ($n = 2,893$) combined below-average temperature, precipitation and wind with above-average humidity and pressure; its mean PM2.5 was 63.50. Cluster 1 was warmer, drier and windier but also high-pressure, with mean PM2.5 54.69. Wetter, lower-pressure clusters 2 and 3 had means of 24.71 and 13.26, respectively.

Table 3: Weather-defined cluster profiles and observed PM2.5.

Cluster	n	Meteorological interpretation	Mean PM2.5
0	2,893	Cooler; humid; drier; high pressure; low wind	63.50
1	1,779	Warmer; drier air; high pressure; higher wind	54.69
2	2,773	Warm; humid; wet; low pressure; cloudy; low wind	24.71
3	192	Warm; humid; wet; low pressure; cloudy; high wind	13.26

3.3 Association rules

Apriori identified 3,753 frequent itemsets and 208 rules leading to high-PM2.5 categories. The strongest retained rule combined medium cloud cover, low temperature and low wind speed and led to Unhealthy PM2.5 with support 0.0206, confidence 0.599 and lift 2.31. High pres-

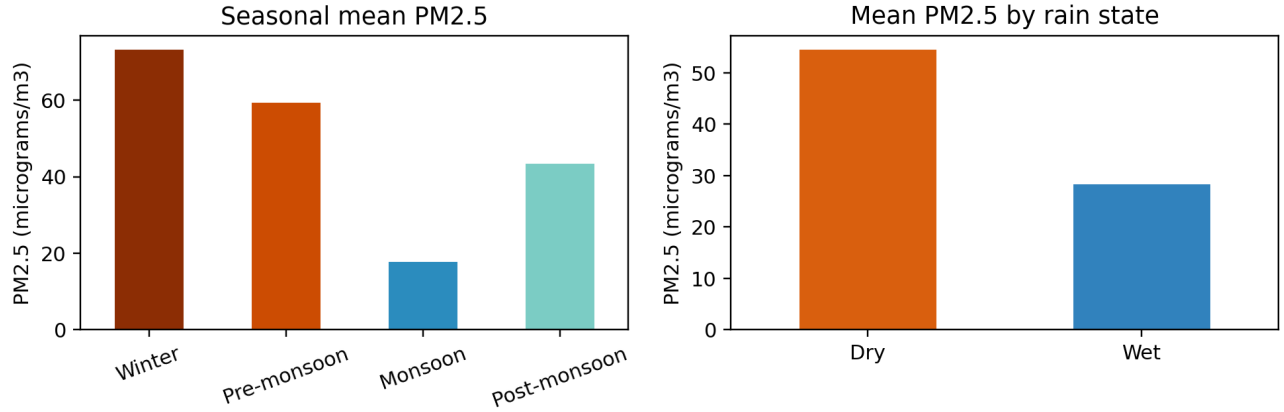


Figure 1: Seasonal and rain-state contrasts in hourly PM2.5. Bars are descriptive means; repeated hourly observations mean they should not be interpreted as independent experimental groups.

sure and low temperature recurred across the leading rules. A broader rule—high pressure, low temperature and southerly wind—covered 10.5% of all observations and had confidence 0.577 and lift 2.22.

Table 4: Five highest-lift rules. All consequents were Un-healthy PM2.5. VPD: vapour-pressure deficit.

Antecedent (weather combination)	Support	Confidence	Lift
Medium cloud; low temperature; low wind	0.0206	0.599	2.31
High pressure; medium humidity; low temperature	0.0541	0.593	2.28
High pressure; low temperature; low wind	0.0390	0.589	2.27
High pressure; low temperature; medium VPD	0.0622	0.581	2.24
High pressure; low temperature; southerly wind	0.1053	0.577	2.22

Support quantifies population coverage, confidence is the conditional frequency of the outcome among matching records, and lift compares confidence with the outcome base rate. The leading rule is strong but uncommon; therefore it should not be presented as a general prediction rule. Rules also share observations and antecedents, so the list is exploratory rather than a set of independent discoveries.

3.4 Predictive performance and temporal generalization

The high-PM2.5 outcome occurred in 29.7% of records, giving a 70.3% majority-class accuracy baseline. Under random splitting, Random Forest achieved the highest ROC-AUC (0.909) and F1 (0.693). Under chronological testing, the largest ROC-AUC was only 0.711 (k-NN), while Logistic Regression had the highest F1 (0.616) through high recall (0.919) but low precision (0.463). This reversal shows that random sampling allows seasonally and temporally similar observations to appear in training and test sets.

Table 5: Random versus chronological holdout performance.

Model	Random AUC	Chronological AUC	Chronological F1
Random Forest	0.909	0.702	0.596
Decision Tree	0.862	0.702	0.611
k-NN ($k = 15$)	0.883	0.711	0.610
Logistic Regression	0.853	0.692	0.616
Naive Bayes	0.835	0.684	0.612

Five-fold rolling validation was still less stable. Logistic Regression had the best mean ROC-AUC (0.663; SD 0.090), followed by k-NN (0.637; SD 0.141). Mean F1 ranged from 0.246 to 0.313, with large fold-to-fold variation. In the confounding analysis, weather-only Random Forest achieved mean AUC 0.593, month-and-hour alone achieved 0.570, and their combination achieved 0.603. Thus calendar variables captured much of the apparent signal and weather added only modest incremental discrimination across seasons.

Table 6: Five-fold expanding-window validation.

Rolling-CV model	Mean AUC	SD	Mean F1	SD
Logistic Regression	0.663	0.090	0.246	0.229
k-NN ($k = 15$)	0.637	0.141	0.280	0.203
Naive Bayes	0.604	0.141	0.273	0.224
Random Forest	0.587	0.192	0.277	0.210
Decision Tree	0.548	0.157	0.313	0.258

Within the chronological Decision Tree, dew point accounted for 72.3% of impurity-based importance, followed by wind gusts (7.5%), pressure (6.6%), cloud cover (4.0%) and relative humidity (3.7%). This aligns qualitatively with the clustering and association results, although correlated predictors can distort impurity importance and the measure does not reveal causal direction.

4 Discussion

Across three descriptive and pattern-mining approaches, high PM2.5 consistently coincided with cool, dry, higher-pressure and weak-dispersion conditions. The monsoon and wet hours had substantially lower concentrations;

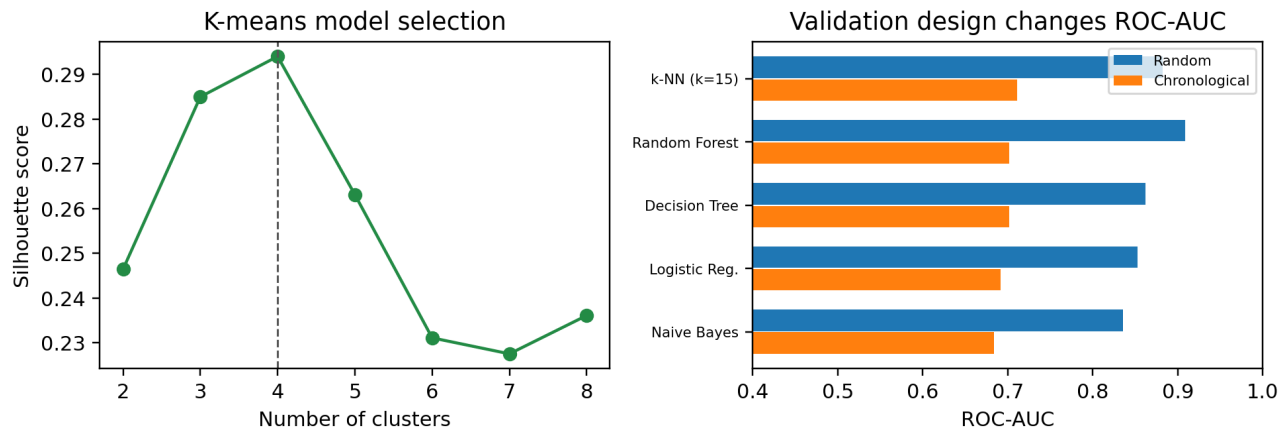


Figure 2: Left: silhouette-based selection of four weather regimes. Right: random splitting produced consistently optimistic ROC-AUC compared with a chronological holdout.

weather-defined wet, low-pressure regimes likewise had the lowest PM_{2.5}. Association rules refined this broad pattern by identifying combinations—notably low temperature with high pressure, low wind or southerly flow—that occurred with Unhealthy concentrations substantially more often than baseline.

These results are physically plausible. Stable winter conditions and a shallow boundary layer limit vertical dilution, while stronger daytime mixing ventilates the basin [2, 3]. Rainfall removes particles and monsoon circulation changes both transport and emissions. Recent multiyear modelling for Kathmandu attributes major variation to seasonal temperature while also identifying biomass burning, long-range transport, diurnal emissions and boundary-layer height [4]. Our weather-only analysis cannot separate those mechanisms, so meteorological patterns should be interpreted as episode contexts rather than emission sources.

The validation experiment is an important caution. Random-split performance appeared strong, but ROC-AUC declined for every classifier when the final portion of the year was held out. Rolling performance was both lower and unstable. Hourly environmental observations are autocorrelated; random splitting can leak seasonal regimes and near-neighbour conditions across the train-test boundary. A model intended for forecasting should therefore be evaluated on future time blocks and, ideally, independent years and sites.

4.1 Implications

For environmental monitoring, the recurring combination of low temperature, elevated pressure and weak wind could support an interpretable warning layer alongside direct PM_{2.5} measurements and numerical forecasts. It should not replace monitoring: the best chronological discrimination was only moderate, and meteorology cannot observe episodic emissions. For policy, the results reinforce the value of winter and pre-monsoon emission controls, when

dispersion is least favourable. Measures could target combustion and resuspended dust sources already documented in Kathmandu [3], while warnings prioritize susceptible populations during stagnant conditions.

4.2 Limitations

First, the dataset covers only one year and appears to represent a single location; generalization across years and the valley is unknown. Second, the maximum PM_{2.5} value of 985 micrograms/m³ occurs in multiple seasons and warrants sensor-quality and provenance checks. Third, hourly observations are not independent, yet clustering and rule support count each hour separately. Fourth, tertile discretization is dataset-relative and may split nearly identical values; different bins or thresholds could alter the rules. Fifth, US EPA categories are defined for 24-hour concentration calculations [6], whereas this study applies their concentration breakpoints to hourly records solely as screening labels. Sixth, multiple overlapping rules were explored without formal false-discovery control. Finally, traffic, household and industrial combustion, fires, boundary-layer height and regional transport were absent; association cannot establish causation.

4.3 Recommended validation

Future work should obtain station metadata and quality flags; aggregate to regulatory-relevant 24-hour means; repeat the analysis across multiple years and sites; use blocked or nested temporal validation; test rule stability by bootstrap or year-wise replication; compare tertiles with domain-based cut points; and add emissions, fire, boundary-layer and back-trajectory variables. Reporting uncertainty intervals and calibrating predicted probabilities would strengthen operational relevance.

5 Conclusion

Analysis of 7,637 hourly observations from Kathmandu in 2019 identified a coherent meteorological signature of elevated PM_{2.5}: cooler, drier, higher-pressure and less ventilated conditions, with substantially cleaner conditions during monsoon and wet hours. K-means separated four overlapping weather regimes, and Apriori produced 208 high-PM_{2.5} rules at prespecified thresholds. The strongest rule had confidence 0.599 and lift 2.31, while a broader high-pressure, low-temperature, southerly-wind rule covered 10.5% of observations with confidence 0.577. Predictive results were more restrained. Strong random-split performance fell to moderate chronological and rolling discrimination, demonstrating that seasonality can make conventional validation optimistic. The study therefore supports interpretable meteorological screening and hypothesis generation, but not causal claims or stand-alone forecasting. Independent temporal and spatial validation is required before operational deployment.

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